El-Houcine Oussi

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Research Interests

Solar power forecasting from minutes to a day ahead; forecast verification and the choice of reference forecasts; probabilistic forecasts and their value in operational decisions such as battery scheduling; satellite imagery and numerical weather prediction as forecast inputs.

Education

M.Sc., Computer Science and Technology Sep 2024 – Jun 2027 (expected)

Nanjing University of Information Science and Technology (NUIST), Nanjing, China

- Research: photovoltaic (PV) power forecasting, model updating, and the calibration of prediction intervals.

State Engineering Degree, Computer Science and Information Systems Sep 2021 – Jun 2024

SUPMTI, Oujda, Morocco

- Coursework included distributed systems, software architecture, database systems and cloud integration.

Specialized Technician Diploma, Software Development Sep 2019 – Jun 2021

ISTA Lazaret, Oujda, Morocco

- Object-oriented programming (Java, C#, C++), relational database design, algorithms, UML and MERISE.

Manuscripts in Preparation

1. Reference specification, model updating and calibration population in hourly photovoltaic power forecasting. *In preparation.* Code and results: doi:10.5281/zenodo.23590205
2. When do probabilistic solar forecasts earn their keep? Forecast quality, distributional validity and operational value in behind-the-meter battery scheduling. *In preparation.*

Software and Data

pv-forecast-reference-evaluation (v1.1.0, MIT licence) 2026

- Code and aggregate results for manuscript 1, archived on Zenodo (10.5281/zenodo.23590205) and on GitHub (github.com/Oussi108/pv-forecast-reference-evaluation).
- Includes a SHA-256 manifest, byte-identical copies of the scripts as run, and an offline check script. Re-running the release reproduced every released table exactly and the figures pixel for pixel.

Research Experience

Graduate Researcher, NUIST, Nanjing Sep 2024 – present

- **Reference forecasts and model updating (2–12 h).** Compared LightGBM, a GRU and a compact Transformer on DKASC Array 1B (Alice Springs), 31 HKUST rooftop systems (Hong Kong) and NREL PVDAQ system 9068 (Colorado). At the Hong Kong testbed, a correctly specified smart-persistence reference had half the mean absolute error of the common "any-daylight" version (0.115 vs. 0.231), which changes how much skill a learned model appears to have.
- **Value of probabilistic forecasts (day-ahead).** Studied when probabilistic forecasts lower costs in home-battery scheduling, using 91,290 simulated scenarios built from 40 Ausgrid households. A scheduler using the full forecast distribution lowered cost for all 40 households.
- **Satellite and weather-forecast inputs (0.5–6 h).** Forecasts for 50 Australian solar farms (AEMO data) and 94 UK virtual farms from Himawari / Meteosat imagery and issued weather forecasts. Final held-out test in progress.
- **Sky-image nowcasting (5–30 min).** Tested whether ground sky images improve on a LightGBM model that uses PV history alone, on Stanford SKIPP'D. Experiments complete: sky images gave no established gain, reported as a negative result.

Methods

- Analysis plans fixed before the final test (with recorded file hashes for the PVDAQ 9068 protocol); held-out test sets scored once.
- Paired bootstrap confidence intervals with calendar blocks; Holm correction across lead times.
- Checksum manifests, pinned package versions and byte-identical as-run scripts for reproducibility.

Engineering Experience

Backend & Mobile Developer Intern, Majal Berkane (Smart City), Berkane, Morocco Mar 2024 – Jul 2024

- Built a Flutter app (MVVM, Riverpod) and REST APIs in ASP.NET Core following Clean Architecture.

Web Development Intern, Provincial Directorate of Equipment, Transport and Logistics, Oujda Mar 2023 – Jul 2023

- Built and deployed an internal Django application for staff records, equipment scheduling and reporting.

Independent developer (SystemX) 2022 – present

- Published MystoGuess, a Flutter guessing game, on Google Play; built Connect Check, a real-time Flutter and Firebase app.

Skills

Research computing	Python, PyTorch, LightGBM, scikit-learn, pandas, NumPy, pvlib, matplotlib, Jupyter, Google Colab
Data	Himawari and Meteosat SEVIRI imagery, numerical weather prediction (GRIB), ERA5, PV plant measurements
Software	C#, ASP.NET Core, Java, Dart / Flutter, Django, SQL, Git, Docker, Linux
Languages	Arabic and Tamazight (native), English (fluent), French (professional), Mandarin Chinese (HSK 3)

References

Available on request.